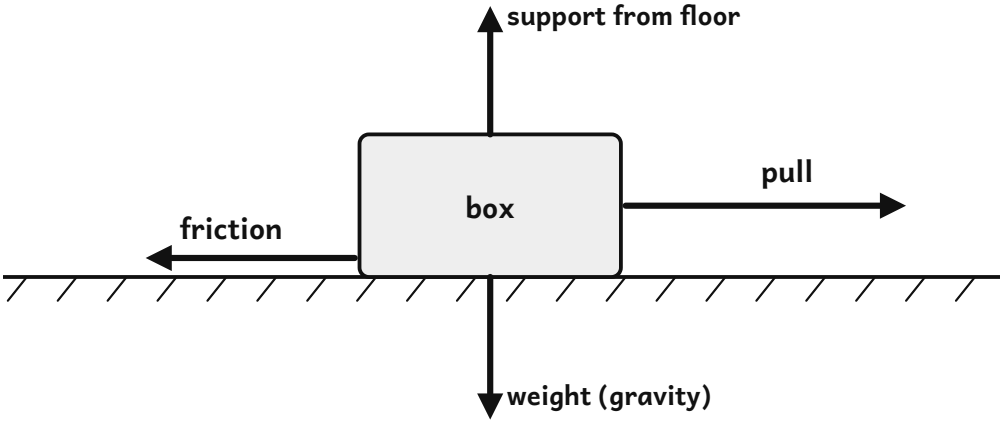
PHYSICAL AND CHEMICAL SCIENCE

Forces and Motion

Pushes and pulls, newtons, mass and weight, friction, gravity and tension, and the laws of motion



In this pack

🕮 **Learn it:** Forces: pushes, pulls and motion ★☆☆
Fact check: Forces around us ★☆☆
Think it through: Forces in action ★★☆
Investigate: Which surface has the most friction? ★★☆
Real-world case study: Kabaddi and tug-of-war: the forces in our games ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Forces: pushes, pulls and motion

Name: Class: Date:

Learning goals

- Name types of force, show their direction with arrows and measure them in newtons.
- Explain the difference between mass (kg) and weight (N).
- Describe the laws of motion and how friction and air resistance affect movement.

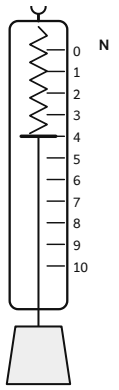
A rickshaw puller pushes the pedals. A goalkeeper stops a football. A coconut falls from a tree. A boat glides across a pond. In every case, a **force** is at work. A force is a **push** or a **pull**. You cannot see forces, but you can see what they do.

What forces can do

- make something **start** moving, or **stop**;
- make it go **faster** or **slower**;
- change its **direction**;
- change its **shape**: squash, stretch, bend or twist (a lump of clay, a spring, a football when kicked).

Measuring forces

Forces are measured in **newtons** (N), named after the scientist Isaac Newton. We measure them with a **forcemeter** (newton meter), which has a spring inside. The bigger the pull, the more the spring stretches.



A forcemeter reading 4 N.

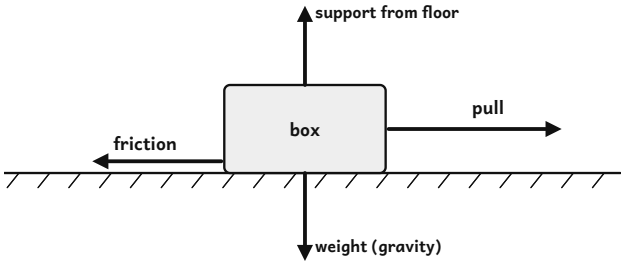
On a diagram we show a force with an **arrow**. The arrow points in the **direction** of the force. A longer arrow means a bigger force.

Mass and weight

Mass is how much matter (stuff) is in an object. It is measured in **kilograms** (kg) with a balance. **Weight** is the **force of gravity** pulling the object down, so it is measured in **newtons** (N). On Earth, a mass of 1 kg has a weight of about 10 N. In everyday life we say "I weigh 30 kg", but scientists would say "my mass is 30 kg".

	Mass	Weight
what it is	amount of matter	pull of gravity
unit	kilogram (kg)	newton (N)
on the Moon	stays the same	about 6 times smaller

Kinds of force



Forces on a box being pulled.

Force	What it does	Contact or not?
gravity	pulls everything down towards the Earth	non-contact
friction	slows things when surfaces rub together	contact
air resistance	friction from the air; slows moving things	contact
tension	the pull in a stretched rope, string or spring	contact
upthrust	water or air pushing up on an object	contact
magnetic force	pulls or pushes magnets and iron	non-contact

Contact forces need the objects to touch. **Non-contact forces** act at a **distance**: gravity pulls a mango down from a tree without touching it.

Friction: friend and enemy

Friction helps us walk without slipping, lets brakes stop a bicycle and stops a rickshaw's tyres sliding. But friction also wastes energy as heat and wears out shoes and tyres. **Air resistance** slows a parachute so it lands gently, but it slows down cyclists and buses too, so racing bicycles are shaped to cut through the air.

The laws of motion

- **1. Things keep doing what they are doing.** A still ball stays still, and a moving ball keeps moving in a straight line, unless a force acts on it. (A ball rolling on grass stops because of friction.)
- **2. Bigger force, bigger change.** Kick a ball harder and it speeds up more. A heavy cart needs a bigger push than a light one.
- **3. Every push has a push back.** When you push the water backwards with an oar, the water pushes the boat forwards. A rocket pushes gas down and the gas pushes the rocket up.

Movement passes from object to object

In carrom, the moving **striker** hits a coin. The striker slows down and the coin moves off: movement energy passes from one object to the other. The faster the striker, the further the coin goes. Moving things have **movement (kinetic) energy**; the faster they go, the more energy they have.

Key words

force	a push or a pull
newton (N)	the unit of force
mass	how much matter is in something (kg)
weight	the pull of gravity on something (N)
gravity	the force that pulls things towards the Earth
friction	a force that slows things when surfaces rub
air resistance	friction from air
tension	the pulling force in a stretched rope or string

Forces around us

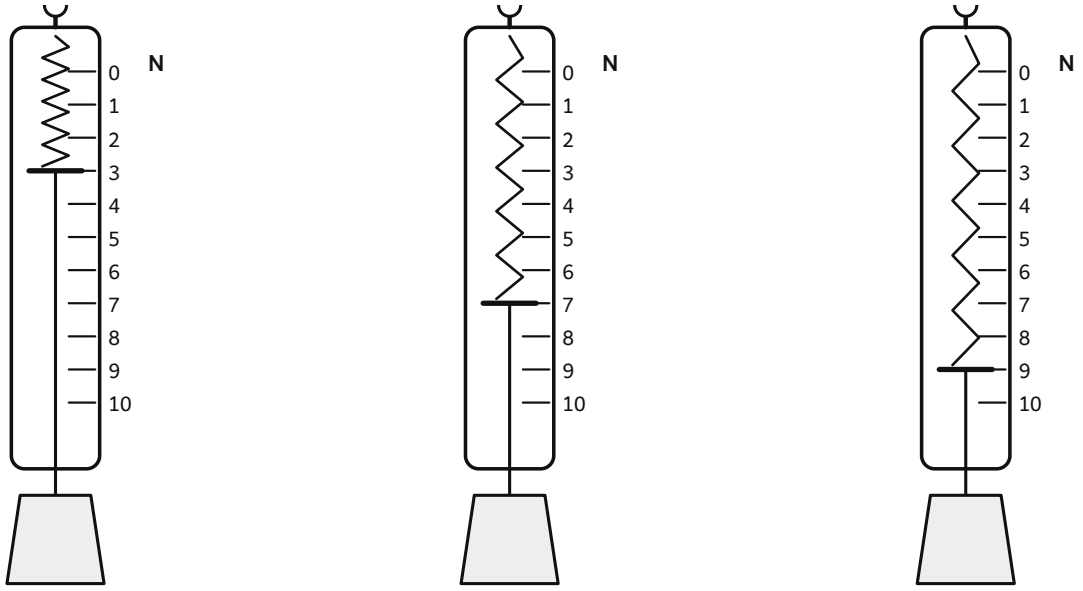
★☆☆ Grade 4

Name: Class: Date:

Learning goals

- Name forces.
- Use units.
- Show forces with arrows.

1 Read the forcemeters. [3]



A *B* *C*

A	B	C

2 Name the force. [5]

What happens	Force
a mango falls from a tree	
a bicycle slows down when you brake	
a rope in tug-of-war is pulled tight	
a parachute falls slowly	
a boat floats	

3 **Sort the forces.** [5]

SORT THESE

friction **a kick** **gravity** **tension** **magnetic force**

Contact force	Non-contact force

4 **Fill in the gaps.** [4]

WORD BANK

smaller **newtons** **bigger** **metres** **kilograms** **same**

Mass is measured in _____. Weight is a force, so it is measured in _____.

On the Moon your mass stays the _____ but your weight is _____.

Forces in action

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Draw force arrows.
- Explain motion using the laws of motion.

1 **Draw arrows to show the forces on a boy on a slide: gravity pulling him down and friction slowing him.** [2]

Slide

2 **Explain each one using a law of motion.** [6]

What happens	Explanation
a football rolling on grass slowly stops	
it is harder to push a full rickshaw van than an empty one	
a swimmer pushes water back and moves forward	

3 **Why do people put sand or gravel on a slippery muddy path in the monsoon?** [2]

4 **In carrom, what happens to the striker's movement when it hits a coin?** [2]

Which surface has the most friction?

★★☆ Grade 4

Name: Class: Date:

Learning goals

- Plan a fair test.
- Measure distance.
- Explain results using friction.

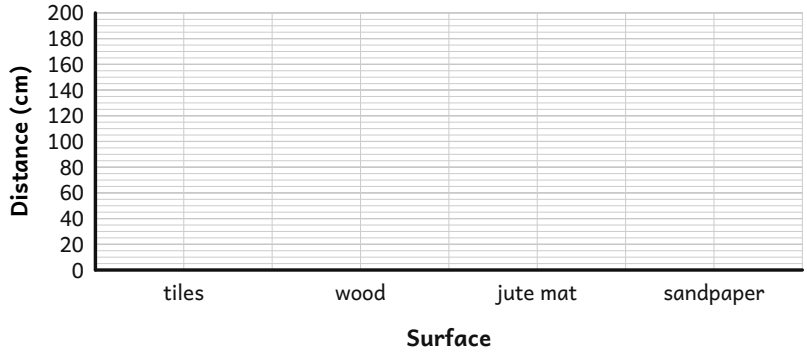
Class 4D let a toy car roll down the same ramp onto four different surfaces and measured how far it travelled before stopping.

1 **Complete the plan.** [3]

Change	
Measure	
Keep the same	

2 **Here are the results (mean of three tries). Draw a bar chart.** [3]

Surface	tiles	wood	jute mat	sand paper
Distance (cm)	180	140	70	35



3 **Which surface had the most friction? How do you know?** [2]

.....

.....

4 **Why did they do each test three times and find the mean?** [1]

.....

5 **Which surface would be best for a playground slide landing area? Why?** [1]

.....

REAL-WORLD CASE STUDY · C4 · FORCES AND MOTION

Kabaddi and tug-of-war: the forces in our games

★★★ Grade 4

Name: Class: Date:

Learning goals

- Find forces in real games.
- Use data from a tug-of-war.
- Think about fairness and safety.

Science · Physical education · Maths · Culture

Kabaddi is the national game of Bangladesh. A raider runs into the other team's half, tries to tag players and get back, while defenders try to hold them. Players **push**, **pull**, twist and change direction quickly. They play barefoot on soil or a mat, which gives good **friction** for gripping the ground.

In **tug-of-war** (*dori tana*), two teams pull a rope in opposite directions. The rope is in **tension**. If both teams pull with the same force, the forces are **balanced** and the rope does not move. If one team pulls harder, the forces are **unbalanced** and the rope moves towards the stronger team.

Example game data below.

1 **Two teams pull a rope (example data).** [4]

Round	Team Shapla (N)	Team Doel (N)	Which way does the rope move?
1	800	800	
2	900	750	
3	700	850	

a) What is the difference in force in round 2?

2 **Why do kabaddi players play barefoot on soil or a mat, not on smooth tiles?** [2]

3 **When a raider pushes off the ground to run, which law of motion is at work? Explain.** [2]

4 **Suggest two rules that make kabaddi or tug-of-war safe and fair.** [2]

Show what you know

★★★
Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

J	T	W	M	O	U	Y	H	F	H
I	V	E	A	F	M	V	J	O	C
C	L	I	S	R	O	J	G	G	L
Y	J	G	S	I	T	T	U	R	F
E	P	H	F	C	I	A	P	A	F
F	I	T	P	T	O	F	U	V	O
C	S	R	U	I	N	E	L	I	R
V	J	D	S	O	T	K	L	T	C
J	L	C	H	N	A	B	Y	Y	E
N	E	W	T	O	N	I	J	H	P

force friction gravity mass motion newton pull push weight

2 Which is a non-contact force? [1]

- ◯ friction
- ◯ tension
- ◯ gravity
- ◯ a push

3 What unit is used for weight? [1]

- ◯ kilograms
- ◯ newtons
- ◯ metres
- ◯ litres

4 Draw a ball being kicked. Use arrows to show the kick, gravity and air resistance. [3]

Kicked ball

5 **Challenge. An astronaut has a mass of 60 kg. What is her mass on the Moon? Is her weight bigger or smaller there? Explain.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
measure forces in newtons	○	○	○
explain mass and weight	○	○	○
name contact and non-contact forces	○	○	○
explain the three laws of motion with examples	○	○	○
investigate friction on different surfaces	○	○	○
explain balanced and unbalanced forces in tug-of-war	○	○	○